Reuben Tamakloe, PhD

Research Assistant Professor

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EDUCATION

University of Seoul — Seoul, South Korea
Doctor of Philosophy in Transportation Engineering (Smart Cities minor) — Sep 2017 – Feb 2022

University of Ghana — Accra, Ghana
Bachelor of Science in Statistics with Mathematics — Aug 2010 – Jul 2014

ACADEMIC APPOINTMENTS

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea
Research Assistant Professor — Sep 2023 - **Present**

New York University (NYU Tandon) — New York, USA
Visiting Postdoctoral Fellow — Jul 2024

University of Seoul — Seoul, South Korea
Research Professor — Apr 2022 – Aug 2023

RESEARCH INTERESTS

Traffic safety, human factors, applications of statistical methods and machine learning/artificial intelligence tools
Autonomous vehicles, electric vehicles
Simulator experiments, optimization, human-centered design
Travel behavior, transport policy

AWARDS

Research Projects/Grants

Awarding Institution	*Fellowship / Research Grant / Conference*	*Grant Amount*	*Duration/Year*
Fellowship			
Korea Advanced Institute of Science and Technology	2025 Jang Young Sil Fellowship	KRW 50,000,000 (~ US$ 34,000)	2025
Grant			
National Research Foundation (NRF; South Korea)	Metaverse-based Testbed Augmentation & Development and Proof of Mixed Traffic Control Strategies (**Co-Investigator**)	KRW 900,000,000 (~ US$ 610,000)	2024-2027
	Exploring the overall efficiency improvement potential of including shared autonomous vehicles in public transit networks (**Principal Investigator**)	KRW 231,700,000 (~ US$ 158,000)	2022-2025
	Development and Validation of a Metaverse-Based Virtual Testing Environment for Autonomous Driving (**Participant**)	KRW 1,800,000,000 (~ US$ 1,200,000)	2023-2024
	Estimating Freight Truck Payloads Using DTG Data to Improve Logistics System Efficiency (**Participant**)	KRW 399,600,000 (~ US$ 270,000)	2022-2023

Awarding Institution	*Fellowship / Research Grant / Conference*	*Grant Amount*	*Duration/Year*
Grant (continued)			
National Research Foundation (NRF; South Korea)	Development of New Transportation Service Solutions Based on a Transportation Platform (**Participant**)	KRW 1,615,000,000 (~ US$ 1,100,000)	2020
Outstanding Paper Presentation Award			
Korea Society of Transportation	88th Academic Conference of the Korean Society of Transportation held at the Korea Science and Technology Hall, South Korea	N/A	2023
Best Paper Award for Graduate Students			
University of Seoul Industry-Academic Cooperation Foundation	N/A	N/A	2022, 2021
Excellent Thesis Support for Graduate Students			
University of Seoul	N/A	N/A	2021
University of Seoul Smart City Department Scholarship			
Smart City Department, University of Seoul	N/A	N/A	2020
Travel grant for conference attendance (ITS World Congress – Singapore)			
National Research Foundation of Korea	N/A	N/A	2019
University of Seoul International Student Scholarship			
University of Seoul	N/A	N/A	2017-2020
University of Seoul International Student Scholarship			
University of Seoul	N/A	N/A	2017-2020
Graduate Student Scholarship			
Dongil Engineering and Consultants Limited	N/A	N/A	2017-2021
University of Ghana Scholarship Recipient			
University of Ghana Student Financial Aid Office	N/A	N/A	2013

TEACHING EXPERIENCE

University of Seoul — Seoul, South Korea

Department of Transportation Engineering

Logistics Management (Undergraduate course)	Spring 2023
Introduction to Transportation Engineering (Undergraduate course - core)	Spring 2023
Big Data Analysis in Transportation (Undergraduate course)	Fall 2022
Advanced Discrete Choice Model I (Graduate course)	Fall 2022

PUBLICATIONS

Published journal articles

*Corresponding author

[54] **Tamakloe, R.**, Caesar, L. D., & *Kim, I. (**2026**). How vehicle-to-grid configurations shape plug-in electric vehicle purchase intentions: Evidence of non-transferable effects. ***Technological Forecasting and Social Change***, 228, 124671. https://doi.org/10.1016/j.techfore.2026.124671

[53] **Tamakloe, R.**, Caesar, L. D., & *Kim, I. (**2026**). Determinants of battery electric vehicle adoption concerns: insights from commercial fleet owners. ***Transportation Research Part A: Policy and Practice***, *205*, 104874. https://doi.org/10.1016/j.tra.2026.104874

[52] ***Tamakloe, R.**, & Caesar, D. L. (2026). Factors associated with incentive-prioritizing electric vehicle buyers. ***International Journal of Sustainable Transportation***. https://doi.org/10.1080/15568318.2026.2619508

[51] ***Tamakloe, R.**, & Lee, H. (**2026**).Variability of Factors Driving Shared Autonomous Vehicle Adoption: A Cross-Group Analysis Based on Local Ride-Sharing Access and Historical Experience. ***International Journal of Sustainable Transportation***. https://doi.org/10.1080/15568318.2026.2623149

[50] ***Tamakloe, R.**, Adanu, E. K., Das, S., & Adoah, B. E. (**2026**). Fatal distracted driving pedestrian-involved crashes in Ghana: Exploring cluster-specific factor associations using cluster correspondence analysis. ***African Transport Studies***, *4*, 100081. https://doi.org/10.1016/j.aftran.2026.100081

[49] Barua, S., Jafari, M., Starewich, M., **Tamakloe, R.**, Alnawmasi, N., & Das, S. (**2026**). Temporal dynamics in barrier-involved crashes: Determining shifts in driving behavior and injury risk across multi-year periods. ***Scientific Reports***. https://doi.org/10.1038/s41598-026-45273-y

[48] Zhang, K., Chen, B., **Tamakloe, R.**, Bai, Y., & *Kim, I. (**2026**). Does the streetscape built environment matter in explaining crash injury severity among older adults?. ***Journal of Transport Geography***, *131*, 104540. https://doi.org/10.1016/j.jtrangeo.2025.104540

[47] Somvanshi, S., Liu, J., Chakraborty, R., **Tamakloe, R.**, & *Das, S. (**2026**). Predicting Crash Severity using Naturalistic Driving Data and Neural Networks. ***International Journal of Intelligent Transportation Systems Research***, 1-18. https://doi.org/10.1007/s13177-025-00624-3

[46] **Tamakloe, R.**, Khorasani, M., *Kim. I. (**2025**). Differences in injury severities between elderly and non-elderly taxi driver at-fault crashes: Temporal instability and out-of-sample prediction. ***Accident Analysis & Prevention***, 211, 107865. https://doi.org/10.1016/j.aap.2024.107865

[45] **Tamakloe, R.**, Zhang, K., Kwon, D., & *Kim, I. (**2025**) Elderly pedestrian safety in silver zones: A temporal instability analysis of injury severity determinants. ***Journal of Traffic and Transportation Engineering***.

[44] **Tamakloe, R**., Khorasani, M., Das, S., & *Kim, I. (**2025**). Pattern recognition in crash clusters involving vehicles with advanced driving technologies. ***Accident Analysis & Prevention***, *218*, 108072. https://doi.org/10.1016/j.aap.2025.108072

[43] *Kutela, B. **Tamakloe, R.**, Novat, N., Kodi, J. H., Kinero, A., Shita, H. (**2025**). Will the widespread driverless passenger vehicles help the elderly and people with disabilities live more independently? Insights from a nationwide survey. ***Cities,*** 159, 105755. https://doi.org/10.1016/j.cities.2025.105755

[42] Antariksa, G., **Tamakloe, R.**, Liu, J., *Das, S. (**2025**). Automated and Explainable Artificial Intelligence to Enhance Prediction of Pedestrian Injury Severity. **IEEE Transactions on Intelligent Transportation Systems**. 10.1109/TITS.2025.3526217

[41] Barua, S., Mimi, M. S., Javed, S. A., **Tamakloe, R.**, & *Das, S. (**2025**). Impact of temporal, spatial, and roadway factors on driver overrides in Level 2 automation: A bivariate binary probit model analysis. ***Transportation Research Part F: Traffic Psychology and Behaviour***, *115*, 103356. https://doi.org/10.1016/j.trf.2025.103356

[40] Jafari, M., Das, S., **Tamakloe, R.**, Khan, M. N., & Hossain, A. (**2025**). Uncovering Individual Heterogeneity in Pedestrian Crash Severity with Mixed Logit Models: A Louisiana Case Study. ***Transportation Research Record.*** *https://doi.org/10.1177/03611981251336135*

[39] Jafari, M., Starewich, M., Das, S., Barua, S., & **Tamakloe, R.** (**2025**). Temporal stability analysis of crash injury severity in school zones: A mixed logit modeling approach. ***IATSS Research***, *49*(4), 528-543. https://doi.org/10.1016/j.iatssr.2025.11.002

[38] ***Tamakloe, R**., & Caesar, D. L. (**2024**). Decoding the patterns of critical factor associations driving electric vehicle recommendations. ***Transportation Research Part A - Policy and Practice.*** 187, 104171. https://doi.org/10.1016/j.tra.2024.104171

[37] ***Tamakloe, R.**, Lee J. K., & Park D. (**2024**). Efficiency variations of integrated urban transit stations before and during the COVID-19 pandemic: Insights from a Mega-City without pandemic lockdown measures. ***Cities***, 152, 105196. https://doi.org/10.1016/j.cities.2024.105196

[36] **Tamakloe, R.**, Zhang, K., *Kim, I. (**2024**). Temporal instability of the determinants of fatal/severe elderly pedestrian injury outcomes in intersections and non-intersections before, during, and after the COVID-19 pandemic. ***Accident Analysis & Prevention***, 205, 107676. https://doi.org/10.1016/j.aap.2024.107676

[35] ***Tamakloe, R.** & Adanu, E. K. (**2024**). Critical Patterns Associated with Vehicle-Pedestrian Hit-and-Run Casualty Injury Severity under Different Weather Conditions: An Association Rule Mining Approach. ***IATSS Research***, 48 (3), 299-318. https://doi.org/10.1016/j.iatssr.2024.06.003

[34] **Tamakloe, R.**, Zhang, K., Hossain, A., Kim, I., & *Park, S. H. (**2024**). Critical risk factors associated with fatal/severe crash outcomes in personal mobility device rider at-fault crashes: A two-step inter-cluster rule mining technique. ***Accident Analysis & Prevention***, 199, 107527. https://doi.org/10.1016/j.aap.2024.107527

[33] ***Tamakloe, R.**, Zhang, K., Atandzi, J., & Park, D. (**2024**). Examining urban delivery service user profiles and determinants of drone delivery adoption in Ghana considering usage before and after the COVID-19 pandemic. ***Transport Policy***, 146, 279-294. https://doi.org/10.1016/j.tranpol.2023.12.004

[32] Caesar, D. L., & ***Tamakloe, R.** (**2024**). Unraveling the patterns of critical contributory factors and flag-state affiliations linked with maritime incident outcomes. ***Marine Policy***, 163, 106135. https://doi.org/10.1016/j.marpol.2024.106135

[31] Oh T., Lim, J., **Tamakloe, R.**, *Li, Z., *Kim, I. (**2024**). Enhancing mutual understanding of e-scooter user's perspective in overtaking maneuver through replaying own driving trajectory. ***Accident Analysis & Prevention***, 207, 107750. https://doi.org/10.1016/j.aap.2024.107750

[30] Zhang, K., **Tamakloe, R.**, *Kim, I. (**2024**). Exploring fatal/severe pedestrian injury crash frequency at school zone crash hotspots: using interpretable machine learning to assess the micro-level street environment. ***Journal of Transport Geography***, 121, 104034. https://doi.org/10.1016/j.jtrangeo.2024.104034

[29] *Adanu, E. K. **Tamakloe, R.**, Dzinyela, R., Agyemang, W. (**2024**). Assessing the factors associated with pedestrian injury severity in hit-and-run crashes in Ghana. ***Transportation Letters***, 1-11. https://doi.org/10.1080/19427867.2024.2366730

[28] Chengula, T. J., *Kutela, B., Novat, N., Shita, H., Kinero, A., **Tamakloe, R.**, Kasomi, S. (**2024**). Spatial instability of crash prediction models: A case of scooter crashes. ***Machine Learning with Applications***, 17, 100574. https://doi.org/10.1016/j.mlwa.2024.100574

[27] ***Tamakloe, R.**, Adanu, E. K., Atandzi, J., Das, S., Lord, D., & Park, D. (**2023**). Stability of factors influencing walking-along-the-road pedestrian injury severity outcomes under different lighting conditions: a random parameters logit approach with heterogeneity in means and out-of-sample predictions. ***Accident Analysis & Prevention***, 193, 107333. https://doi.org/10.1016/j.aap.2023.107333

[26] *Das, S., Dutta, A., **Tamakloe, R.**, & Khan, M. N. (**2023**). Analyzing the time-varying patterns of contributing factors in work zone-related crashes. ***Journal of Transportation Safety & Security***, 16 (6), 655-682. https://doi.org/10.1080/19439962.2023.2246020

[25] ***Tamakloe, R.**, Das, S., Adanu, E. K., & Park, D. (**2023**). Key factors affecting motorcycle-barrier crash severity: an innovative cluster-regression technique. ***Transportmetrica A: Transport Science***. 21 (1), 2230310. https://doi.org/10.1080/23249935.2023.2230310

[24] ***Tamakloe, R**., & Park, D. (2023). Discovering latent topics and trends in autonomous vehicle-related research: A Structural Topic Modelling approach. ***Transport Policy***, 139, 1-20. https://doi.org/10.1016/j.tranpol.2023.06.001

[23] ***Tamakloe, R.** (**2023**). Risk Factors Influencing Fatal Powered Two-Wheeler At-Fault and Not-at-Fault Crashes: An Application of Spatio-Temporal Hotspot and Association Rule Mining Techniques. ***Informatics***, 10 (2), 43. https://doi.org/10.3390/informatics10020043

[22] *Das, S., **Tamakloe, R.**, Zubaidi, H., Obaid, I., & Rahman, M. A. (**2023**). Bicyclist injury severity classification using a random parameter logit model. ***International Journal of Transportation Science and Technology***, 12 (4), 1093-1108. https://doi.org/10.1016/j.ijtst.2023.02.001

[21] *Zubaidi, H., **Tamakloe, R.**, Al-Bdairi, N. S. S., *Alnedawi, A., & Obaid, I. (**2022**). Exploring senior motorcyclist injury severity crashes: Random parameter model with heterogeneity in mean and variance. ***IATSS Research***, 47 (1), 1-13. https://doi.org/10.1016/j.iatssr.2022.12.001

[20] **Tamakloe, R.**, & *Park, D. (**2022**). Factors influencing fatal vehicle-involved crash consequence metrics at spatio-temporal hotspots in South Korea: application of GIS and machine learning techniques. ***International Journal of Urban Sciences***, 27 (3), 483-517. https://doi.org/10.1080/12265934.2022.2134182

[19] **Tamakloe, R.**, Park, D., & *Chang, H. (**2022**). Discovering research topics, trends, and perspectives in COVID-19-related transportation journal articles. ***International Journal of Urban Sciences***, *26*(4), 710-738. https://doi.org/10.1080/12265934.2022.2044891

[18] **Tamakloe, R.**, Hong, J., Kim, J., & *Park, D. (**2022**). Factors affecting fatal PTW at-fault crash outcome metrics at intersections and non-intersections in South Korea. ***Journal of Transportation Safety & Security***, 15 (7), 681-716. https://doi.org/10.1080/19439962.2022.2123582

[17] *Das, S., **Tamakloe, R.**, Kutela, B., & Hossain, A. (**2022**). Pattern recognition from injury severity types of frontage roadway crashes. ***Journal of Transportation Safety & Security***, 15 (7), 659-680. https://doi.org/10.1080/19439962.2022.2123581

[16] **Tamakloe, R.**, Sam, E. F., Bencekri, M., Das, S., & *Park, D. (**2022**). Mining groups of factors influencing bus/minibus crash severities on poor pavement condition roads considering different lighting status. ***Traffic injury prevention***, *23*(5), 308-314. https://doi.org/10.1080/15389588.2022.2066658

[15] *Obaid, I., Alnedawi, A., Aboud, G. M., **Tamakloe, R.**, Zubaidi, H., & Das, S. (**2022**). Factors associated with driver injury severity of motor vehicle crashes on sealed and unsealed pavements: random parameter model with heterogeneity in means and variances. ***International journal of transportation science and technology***, 12 (2), 460-475. https://doi.org/10.1016/j.ijtst.2022.04.002

[14] **Tamakloe, R.**, Das, S., Aidoo, E. N., & *Park, D. (**2022**). Factors affecting motorcycle crash casualty severity at signalized and non-signalized intersections in Ghana: Insights from a data mining and binary logit regression approach. ***Accident Analysis & Prevention***, *165*, 106517. https://doi.org/10.1016/j.aap.2021.106517

[13] *Das, S., **Tamakloe, R.**, Zubaidi, H., Obaid, I., & Alnedawi, A. (**2021**). Fatal pedestrian crashes at intersections: Trend mining using association rules. ***Accident Analysis & Prevention***, *160*, 106306. https://doi.org/10.1016/j.aap.2021.106306

[12] **Tamakloe, R.**, Lim, S., Sam, E. F., Park, S. H., & *Park, D. (**2021**). Investigating factors affecting bus/minibus accident severity in a developing country for different subgroup datasets characterised by time, pavement, and light conditions. ***Accident Analysis & Prevention***, *159*, 106268. https://doi.org/10.1016/j.aap.2021.106268

[11] **Tamakloe, R.**, Hong, J., Tak, J., & *Park, D. (**2021**). Finding evacuation routes using traffic and network structure information. ***Transportation research part D: transport and environment***, *95*, 102853. https://doi.org/10.1016/j.trd.2021.102853

[10] **Tamakloe, R.**, Hong, J., & *Park, D. (**2020**). A copula-based approach for jointly modeling crash severity and number of vehicles involved in express bus crashes on expressways considering temporal stability of data. ***Accident Analysis &***

Prevention, *146*, 105736. https://doi.org/10.1016/j.aap.2020.105736

[9] Hong, J., **Tamakloe, R.**, & *Park, D. (**2020**). Application of association rules mining algorithm for hazardous materials transportation crashes on expressway. ***Accident Analysis & Prevention***, *142*, 105497. https://doi.org/10.1016/j.aap.2020.105497

[8] Hong, J., **Tamakloe, R.**, Tak, J., & *Park, D. (**2020**). Two-Stage Double Bootstrap Data Envelopment Analysis for Efficiency Evaluation of Shared-Bicycle Stations in Urban Cities. ***Transportation Research Record***, *2674*(6), 211-224. https://doi.org/10.1177/0361198120918568

[7] Hong, J., **Tamakloe, R.**, & *Park, D. (**2020**). Discovering insightful rules among truck crash characteristics using apriori algorithm. ***Journal of advanced transportation***, *2020*, 1-16. https://doi.org/10.1155/2020/4323816

[6] **Tamakloe, R.**, *Hong, J., & Tak, J. (**2021**). Determinants of transit-oriented development efficiency focusing on an integrated subway, bus and shared-bicycle system: Application of Simar-Wilson's two-stage approach. ***Cities***, *108*, 102988. https://doi.org/10.1016/j.cities.2020.102988

[5] **Tamakloe, R.**, & *Hong, J. (**2020**). Assessing the efficiency of integrated public transit stations based on the concept of transit-oriented development. ***Transportmetrica A Transport Science***, *16*(3), 1459-1489. https://doi.org/10.1080/23249935.2020.1753849

[4] Hong, J., **Tamakloe, R.**, Lee, S., & *Park, D. (**2019**). Exploring the topological characteristics of complex public transportation networks: focus on variations in both single and integrated systems in the Seoul metropolitan area. ***Sustainability***, *11*(19), 5404. https://doi.org/10.3390/su11195404

[3] Hong, J., **Tamakloe, R.**, & *Park, D. (**2019**). A comprehensive analysis of multi-vehicle crashes on expressways: a double hurdle approach. ***Sustainability***, *11*(10), 2782. https://doi.org/10.3390/su11102782

[2] Hong, J., **Tamakloe, R.**, Lee, G., & *Park, D. (**2019**). Insight from scientific study in logistics using text mining. ***Transportation research record***, *2673*(4), 97-107. https://doi.org/10.1177/0361198119834905

[1] Hong, J., **Tamakloe, R.**, *Park, D., & Choi, Y. (**2019**). Estimating incident duration considering the unobserved heterogeneity of risk factors for trucks transporting HAZMAT on expressways. ***Transportation research record***, *2673*(2), 232-242. https://doi.org/10.1177/0361198119827925

Selected journal papers under review (SCI(E))

*Corresponding author

[7] **Tamakloe, R.**, Caesar, L. D., & *Kim, I. **(2026).** Age Heterogeneity in Willingness to Use Empty Self-Driving Vehicles for Child Transport. ***Transportation Research Part A***

[6] **Tamakloe, R.**, Adanu, E. K., Sam, E. F., & *Kim, I. (2026). Cluster-Specific Risk Factor Patterns Associated with Fatal/severe Injury Rural Bus/minibus Crashes in a Low- and Middle-Income Country. ***Accident Analysis & Prevention.***

[5] Chun, G., **Tamakloe, R.**, & *Kim, I. **(2026).** Route Choice Behavior Analysis of Electronic Scooter Users: Insights from Built Environment Features Using Street View Image. ***Cities***

[4] **Tamakloe, R.**, Yoo, Y., Zhang, K., *Kim. I. (2026). Value-based adoption of electric vehicles: An empirical analysis of continued usage amid rising battery fires in South Korea, ***Research in Transportation Business & Management.***

[3] **Tamakloe, R.**, Zhang, K., *Kim, I. (2026). Assessment of Micro-Scale Built Environment Measures and Temporal Instability in Pedestrian Injury Severity Outcomes in School and Non-School Zones. ***International Journal of Injury Control and Safety Promotion.***

[2] **Tamakloe, R.**, Zhang, K., *Kim, I. (2025). Unraveling Seasonal Variations in Factors Impacting Pedestrian Injury Severity in Hit-and-Run Crashes. ***Travel Behaviour and Society.***

[1] ***Tamakloe, R.** & Adanu, E. K. (2025). Insights from Injury Analysis on Passenger Falls from Moving Vehicle Incidents: An Association Rules Mining Approach. ***IATSS Research***

CONFERENCE PROCEEDINGS/INVITED LECTURES

Invited Guest Lecturer

[4] **UN Economic and Social Commission for Asia and the Pacific**, *Daejeon, South Korea* (3 December 2025); Integrating Human-Centered Safety into Sustainable Mobility Systems.

[3] **New York University**, *New York, USA* (25 July 2024); Leveraging artificial intelligence to improve safety for personal mobility device users.

[2] **New York University**, *New York, USA* (23 July 2024); Can we improve the efficiency of public transport systems by integrating shared autonomous shuttles into them?

[1] **Korea Advanced Institute of Science and Technology**, *Daejeon, South Korea* (12 October 2023); Discovering the factors influencing shared autonomous vehicle adoption: A cross-group analysis considering local ride-sharing access and historical usage experience.

International Presentations

The TRB 104th Annual Meeting - Transportation Research Board Annual Meeting, *Washington DC, USA* (5-9 Jan. 2025)

[20] Elderly Pedestrian Crash Severity: Exploring the Role of Human Perceived Streetscape Design Using Explainable Ensemble Learning

[19] Identifying Patterns and Risk Factors in SUV-Related Pedestrian Crashes Using Cluster Correspondence Analysis

ICTCT 2024, *Hague, Netherlands* (17-18 Oct. 2025)

[18] Factors Influencing Injury Severity Among Elderly Pedestrians - Insights from Urban Imagery and Explainable Machine

Learning

[17] Differences between pedestrian injury severity determinants at school and non-school zones: new findings from streetscape quality factors

[16] Variability in the impact of factors influencing elderly pedestrian injuries in crashes at intersections and non-intersections before, during, and after the COVID-19 pandemic

[15] The Role of Human Subjective Perceptions and Objective Measurements of the Urban Environment in Explaining Pedestrian Injury Severity

The TRB 103rd Annual Meeting - Transportation Research Board Annual Meeting, *Washington DC, USA* (7-11 Jan. 2024)

[14] Investigating the Performance of Integrated Transit Stations: A Comparative Study in a Mega-City Before and During the COVID-19 Pandemic

[13] Discovering the Factors Influencing Shared Autonomous Vehicle Adoption: A Cross-Group Analysis Considering Local Ridesharing Access and Historical Experience

[12] Understanding Consumers Prioritizing Incentives in Electric Vehicle Purchases: Identifying Patterns for Effective Policy Strategies

[11] Uncovering Individual Heterogeneity in Pedestrian Crash Severity with Mixed Logit Models

[10] Investigating Critical Factor Associations Affecting Injury Severity Outcomes of Passengers Falling from Moving Vehicles

[9] Investigating Crashes Occurred at School Zones Using Random Parameter Ordered Probit Model

[8] Patterns of Critical Factors Linked to Automated Vehicle–Involved Crashes: A Comparative Analysis of Intersection and Non-Intersection Crash Scenarios

The TRB 102nd Annual Meeting - Transportation Research Board Annual Meeting, *Washington DC, USA* (8-12 Jan. 2023)

[7] Qualitative Analysis of Urban Air Mobility as a Disruptive Technology

[6] Perceptions of Drone Delivery Adoption in Ghana Considering Online Delivery Usage Before and After the Incidence of the COVID-19 Pandemic

[5] Risk Factors Affecting Motorcycle-Barrier Crashes and Injury Severities: Insights from an Innovative Cluster-Regression Technique

The TRB 101st Annual Meeting - Transportation Research Board Annual Meeting, *Washington DC, USA* (9-13 Jan. 2022)

[4] Investigating Chains of Risk Factors Influencing Fatal Powered Two-Wheeler Crashes at Spatio-Temporal Hotspot Locations in South Korea

[3] Investigating Chains of Factors Influencing Motorcycle Crash Casualty Severity at Signalized and Non-Signalized Intersections in a Developing Country

26th ITS World Congress, Suntec Conference Center, *Suntec, Singapore* (21-25 Oct. 2019)

[2] Assessing the impacts of Land Use on Subway Ridership: Identifying a Suitable Sustainable Transport Policy

The 13th UC-US-KU-TU International Joint Seminar, China and France Center, Tongji University, *Shanghai, China* (24 Aug. 2018)

[1] An Analysis of Bus-involved Crashes on Expressway through Conditional Mixed Process

Local (South Korea)

****Student supervisions***

[23] *Comparing Temporal Instability in Injury Severities Between Elderly and Non-Elderly Taxi Drivers in At-Fault Crashes (2024.10). Korean Institute of ITS Conference, Jeju, South Korea

[22] *Analyzing Risk Factor Patterns in Homogeneous Crash Groups: A Comparative Study of ADAS and ADS (2024.4). Korean Institute of ITS Conference, Jeju, South Korea

[21] Factors Influencing Light-Duty Full Electric Vehicle Adoption among Companies (2024.4). Korean Institute of ITS Conference, Jeju, South Korea

[20] * Analysis of subway station operation efficiency and review of environmental factors impact: Focusing on Seoul Metropolitan City (2023.11). Korean Institute of ITS Conference, Jeju, South Korea.

[19] * Review of previous research on passenger-logistics integrated transportation system and suggestion of future research direction (2023.11). Korean Institute of ITS Conference, Jeju, South Korea.

[18] Road lighting condition, pedestrian maneuver and pedestrian injury severity: A joint analysis using an advanced econometric approach (2023.04). Korean Institute of ITS Conference, Yeosu, South Korea.

[17] Are electric vehicle users willing to recommend their vehicles to others? (2023.04). Korean Institute of ITS Conference, Yeosu, South Korea.

[16] * Investigating how usage experience impacts consumer preference variability for shared autonomous vehicles (2023.04). Korean Institute of ITS Conference, Yeosu, South Korea.

[15] * Learning about the variations in research interests regarding last-mile logistics among OECD and Non-OECD countries (2023.04). Korean Institute of ITS Conference, Yeosu, South Korea.

[14] * Hybrid delivery system: A literature review and future directions (2023.04). Korean Institute of ITS Conference, Yeosu, South Korea.

[13] * Trends and patterns in sustainable transportation research (2023.04). Korean Institute of ITS Conference, Yeosu, South Korea.

[12] * The determinants of factors affecting transit station efficiency before, during and after the COVID-19 pandemic (2023.04). Korean Institute of ITS Conference, Yeosu, South Korea.

[11] A structural topic modeling approach to exploring autonomous vehicle-related research (2023.02). The Korean Society of Transportation Conference, Gangnam, Seoul, South Korea.

[10] * Variations in the determinants of bicycle-transit station efficiency pre- and post- COVID-19 pandemic (2023.02). The Korean Society of Transportation Conference, Gangnam, Seoul, South Korea.

[9] Drone delivery for urban logistics after COVID: How can we bolster its adoption? (2022.11). Korean Institute of ITS Conference, Jeju, South Korea.

[8] Performance of public transport stations through the pandemic: Are there any key patterns? (2022.11). Korean Institute of ITS Conference, Jeju, South Korea.

[7] How different are researcher perspectives regarding COVID-19 and transportation? (2022.06) Korean Institute of ITS Conference, Jeju, South Korea.

[6] A machine learning approach to identifying factors influencing motorcycle crashes at critical hotspot locations (2022). KITS International Conference, Jeju JDC Conference Center, Jeju, South Korea.

[5] Finding evacuation paths considering dynamic spatio-temporal vulnerability information and network structure (2020.11). The Korean Institute of ITS Conference, Online.

[4] An Application of Simar-Wilson Bootstrap Data Envelopment Analysis to Investigate the Efficiency of Shared-Bicycle Stations in Seoul (2019.11). The Korean Institute of ITS Conference, St. John's Hotel, Gangneung, South Korea.

[3] Topological Analysis of Multimodal Transportation Network in Seoul Based on Graph-Based Methods (2019.04). The Korean Institute of ITS Conference, Halla Convention center, Cheju Halla University, Jeju, South Korea.

[2] Comparing the characteristics of multivehicle crashes using Cragg's double-hurdle regression model (2018.11). The Korean Institute of ITS Conference, The Genesis Building, Kyungsung University, Busan, South Korea.

[1] A Study for Conditional Mixed Process Model of Bus-related Crashes on Expressway (2018.04). The Korean Institute of ITS Conference, Halla Convention center, Cheju Halla University, Jeju, South Korea.

ANALYTICAL TOOLS

General programming: Python, R
Statistical analysis: STATA, LIMDEP/NLOGIT
Network analysis: Gephi
Geography: Geographic Information Systems (GIS)
Simulation/simulator: Vissim, Unity (learning)

MENTORING

Korea Advanced Institute of Science and Technology (KAIST) – Nano Transportation Research Team

PhD Students
Hyunchul Park (August 2025 to Present)
Kaihan Zhang (September 2023 to Present)
Mahdi Khorasani (September 2023 to Present)

Master's Students
Kexin Zhu (August 2025 to Present)
Myeonghyeong Kim (August 2025 to Present)
Junyoung Kim (August 2025 to Present)
Chanwoo Moon (August 2025 to Present)
Jungyoon Kim (August 2025 to Present)

University of Seoul

PhD Students
Joongkwon Lee (March 2022 to July 2023)

Master's Students
Juliana Ohenewaa Ofosu (February 2023 to February 2025; Currently at Ministry of Transport - Ghana)
Princella Gababor (February 2022 – August 2024; Currently at Ministry of Transport - Ghana)
Yang Hun (March 2022 to July 2023; Currently PhD student at the Seoul National University)
Herin Lee (March 2022 to July 2023)
Situma Phionah (February 2022 – February 2023; Currently at Ministry of Transport - Uganda)
Beverlyn Adoah (February 2021 – February 2023)
Atsu Akpakli (February 2020 – February 2022; Currently at Ministry of Transport - Ghana)

ACTIVE REVIEWING SERVICE

Reliability Engineering & System Safety
Sustainable Cities and Society
Accident Analysis and Prevention
Transport Policy
Research in Transportation Business & Management
Cities
Transportation Research Part A: Policy and Practice
Transportation Research Part D: Transport and Environment
Transportation Research Part E: Logistics and Transportation Review
Transportation Research Part F: Traffic Psychology and Behaviour
Journal of Transport Geography
Travel Behaviour and Society
Case Studies on Transport Policy
Transportation Research Interdisciplinary Perspectives
IATSS Research
Transportmetrica A: Transport Science
Traffic Injury Prevention
Transportation Research Record: Journal of the Transportation Research Board
KSCE Journal of Civil Engineering
International Journal of Urban Sciences
International Journal of Sustainable Transportation

ACADEMIC MEMBERSHIPS

Korean Transportation Association
Ghana Transportation Professionals Forum
KSCE CELeN Student
Korean Society of Hazard Mitigation
Korea Institute of Intelligent Transport Systems

VOLUNTEER AND EXTRA-CURRICULAR ENGAGEMENTS

Operations Committee Member – The 15th Asia-Pacific Conference on Transportation and the Environment, Jeju Island, South Korea (https://apte2026.kr/committee)	July 8-10, 2026
New Student Mentoring Volunteer – International Student Office, Univ. of Seoul	Jan 2021 – Jun 2021
Organizing Secretary – Ghanaian Students in Korea and Associates, South Korea	2020 – Present
University Representative (University of Seoul) – Ghanaian Students in Korea and Associates, South Korea	2019 – 2020
Leader, Program Planning Team – Ghana Association of Statistics Students, University of Ghana	2013 – 2014